EEG-fNIRs integration for brain computer interfaces.

A re-examination of the Beer Lamberts Law

1.0 Abstract:

Electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS) have each emerged as non-invasive, low-cost tools for monitoring brain activity. EEG captures millisecond-scale electrical oscillations, while fNIRS detects slower hemodynamic changes associated with neural metabolism. Their integration is increasingly proposed for brain computer interfaces (BCIs), yet the mathematical foundations that allow meaningful fusion remain underexplored outside of specialist optics literature. Therefore, this paper revisits the Modified Beer Lambert Law (MBLL), the analytical core of fNIRS, and argues that a deeper mathematical appreciation of MBLL is necessary for advancing hybrid EEG+fNIRS systems. We will explore how MBLL adapts the classical Beer Lambert law to scattering tissue, derive its chromophore separation formalism, and critique its assumptions. Finally we propose a novel theoretical extension that incorporates time varying scattering as a measurable signal, potentially enriching EEG+fNIRS fusion beyond existing paradigms.

2.0 Introduction:

Brain activity is multifaceted; neurons communicate via electrical impulses, but sustain themselves through oxygen and glucose metabolism. This duality has led to the need for a multimodal recording method. EEG measures the electrical "input–output" side of neural communication, while fNIRS captures vascular "support" signals tied to oxygen delivery and consumption. When integrated these modalities form a causal chain: neural firing → metabolic demand → vascular response → optical absorption.

Yet, while EEG is underpinned by straightforward electrophysiological laws (superposition of dipole fields), the optical side requires subtler physics. Unlike photons in clear media, light in tissue scatters, refracts, and diffuses, obscuring simple Beer Lambert interpretations. Thus, the Modified Beer Lambert Law (MBLL) was introduced as the practical bridge between absorption physics and usable fNIRS signals.

In this paper, we argue that a rigorous understanding of MBLL is not just academic but critical: only by knowing its assumptions and limitations can EEG and fNIRS be effectively combined. We proceed by deriving MBLL from first principles, critiquing its approximations and suggesting an extensions to its applicability.

3.0 From Beer Lamberts Law to Modified Beer Lamberts Law:

3.1 Classic Beer Lamberts Law:

Let us imagine that light is travelling through a homogenous transparent medium. In every small length of distance dL travelled by the photon, some of its energy (dI) gets absorbed.
Mathematically:

$$dI = -\mu_a I dL$$

Where:
$I = \text{the intensity of the photon at that point}$
$\mu_a = \text{absorption constant (extinction coefficient} \times \text{concentration)}$
The negative sign implies that the intensity decreases as the distance travelled through the medium increases.

Evidently the above derived equation is in the form of a first order homogenous differential equation that can be solved as follows:
Rearranging we get

$$\frac{dI}{I} = -\mu_a dL$$

Taking the integral on both sides over a finite path length yields

$$\int_{I_0}^{I} \frac{dI}{I} = -\mu_a \int_{0}^{L} dL$$

$$ln(\frac{I}{I_0}) = -\mu_a L$$

Removing the logarithm

$$\frac{I}{I_0} = e^{-\mu_a L}$$

$$I = I_0 e^{-\mu_a L}$$

3.2 Challenges with our medium being human tissue

Unlike a homogenous transparent medium, brain tissue scatters and diffracts light strongly thus complicating photon trajectories. As a result:

1. Effective path length required for predictive mathematics exceeds the source detector distance
2. Absorption arises from multiple chromophores (HbO, HbR, water, Lipids)

3. Varying geometries of the skin and skull introduced static distortions.

3.3 Resolving challenge 2

Now since light absorption depends on how much the medium's molecules interact with the photons. This interaction depends on photon energy which is related to its wavelength by

$$E = \frac{hc}{\lambda}$$

Therefore absorption probability is dependent on the wavelength of light.

Hence in the differential equation we derived we need to consider wavelength dependency of the decay coefficient. a the molar extinction coefficient is not universal but changes with wavelength thus targeting the problem that chromophores absorb light differently depending on the energy of the photon.

Therefore to be accurate we modify classical Beer Lambert's Law to be:

$$\frac{dI(\lambda,L)}{I(\lambda,L)} =- \mu_t(\lambda)dL$$

Where:

μ_t is the total attenuation coefficient at wavelength λ

$I(\lambda, L)$ is the intensity at a position L along the photon path

If the medium were purely absorbing and non-scattering then $\mu_t(\lambda) = \mu_a(\lambda)$ and integrating would give us the classic Beer Lamberts Law with wavelength dependence

$$I(\lambda, L) = I_0(\lambda)e^{-\mu_a(\lambda)L}$$

3.4 Justification for the addition of wavelength for fNIRs detection:

When we add wavelength dependency the following is implied:

- Hemoglobin absorbs light differently at different wavelengths
- Through the specificity of wavelength we can exploit the distinct extinction spectra of oxy hemoglobin and deoxy hemoglobin in order to resolve multiple chromophores simultaneously through the modified Beer Lambert's Law.

3.5 Resolving challenge 1 and 3:

In order to resolve the effective path length problem and take into consideration static distortion we will further modify the derived Beer Lamberts Law in the following ways:

1) Adding a differential path length factor thereby scaling the source detector distance to an effective average path.
2) Adding a time dependent difference formulation factor that cancels the peaks and troughs in static distortions and scattering over time.

3.5.1 Introducing the effects of scattering:

In real tissue photons are scattered many times. Scattering does not necessarily imply the elimination of photons as absorption does but it changes its direction so they are less likely to be detected along the original line of incidence thus creating a random error.

To account for this we separate

$$\mu_t(\lambda) = \mu_a(\lambda) + \mu_s(\lambda)$$

In tissues photons predominantly scatter forward. This means that a photon may scatter multiple times but still keep moving in about the same direction as the original direction. If we simply us $\mu_s(\lambda)$ that gives us the number of scattering events per mm we would be overestimating how much scattering randomizes and deviates the photon from its original trajectory. What really matters during diffusion is not just how often photons scatter but more importantly how much their direction changes per scatter. Hence we will be using the following reduced scattering coefficient:

$$\mu'_s(\lambda) = \mu_s(\lambda)(1 - g(\lambda))$$

Where $g(\lambda)\ is\ the\ anisotropy\ factor$

3.5.1.1 Justification for the reduced scattering coefficient and anisotropy factor:

By definition

$$g(\lambda) = cos\theta$$

Where θ is the scattering angle between the incoming and outgoing photon direction

- If $g = 1$ all scattering is perfectly forwards ($\theta = 0$) so scattering barely changes photon direction.
- If $g = 0$ all scattering is isotropic i.e. the photon is equally likely to scatter in any direction.
- If $g =- 1$ the photon scatters perfectly backward.

In biological tissue $g \approx 0.8 - 0.95$ meaning it is highly forward peaked thus justifying our need to use a reduced scattering coefficient $\mu'_s(\lambda)$

3.5.1.2 derivation for $\mu'_s(\lambda)$:

For photons propagating through tissues every scattering event changes the photons direction by some angle θ. the average projection of the new photon direction relative to the old direction is $cos\theta$. Therefore on average after one scattering event the photons momentum will be reduced by $1 - g$. Thus the effective scattering per unit length is not $\mu_s(\lambda)$ but $\mu'_s(\lambda)$.

3.5.2 Diffusion theory and Radiative transfer equation:

If one keeps the simple exponential form with $\mu_t(\lambda)$ we would be incorrectly treating scattering as loss as explained in ***section 3.5.1***. Instead, because scattered photons can still reach the detector after a longer random path, the correct large-scale description of attenuation in a highly scattering medium is governed by diffusion theory (the diffusion approximation to the Radiative Transfer Equation).

The most fundamental model of photon transport in tissue is the radiative transfer equation (RTE), which governs the specific intensity $L(r, s, t)$. This quantity represents radiant power per unit area, per unit solid angle, at position r in direction s, and time t. However the RTE is far too complex and is beyond the scope of this paper.

However through the RTE we know that after many effective scatterings the photon begins to randomize its behaviour and can therefore be modelled by the modified Beer Lamberts Law which is the crux of this exploration.

The Beer Lambert's Law for a homogeneous absorbing medium as determined earlier is:

$$I(\lambda, L) = I_0(\lambda)e^{-\mu_a(\lambda)L}$$

Rearranging

$$\frac{I(\lambda,L)}{I_0(\lambda)} = e^{-\mu_a(\lambda)L}$$

Taking the natural logarithm on both sides gives absorbance (optical density) on the left and yield a linear equation:

$$A(\lambda, L) =- \mu_a(\lambda)L$$

Now we can track small changes in absorption relative to the baseline.$(\mu_a + \Delta\mu_a)$

$$A(\lambda, L) = L(\mu_a + \Delta\mu_a) - \mu_a(\lambda)L$$

Hence

$$\Delta A(\lambda, L) = \Delta\mu_a(\lambda)L$$

This linearized form is the foundation of the Modified Beer Lambert Law as the log transform of intensity and the use of small increments around baseline together justify the linear relationship between ΔA and $\Delta\mu_a$.

In cranial tissue photons do not travel straight across the source–detector separation d they scatter and take tortuous paths, making their average pathlength ⟨L⟩ longer than d.To account for this discrepancy, we introduce the **differential pathlength factor (DPF):**

$$\langle L\rangle_\lambda = d \cdot DPF(\lambda)$$

This correction is essential as without it the linear relation derived from Beer Lambert would underestimate absorption changes in scattering tissue.

Substituting the effective pathlength into the linear absorbance relation gives:

$$\Delta A(\lambda, L) = d \cdot DPF(\lambda) \cdot \Delta\mu_a(\lambda)$$

In the near-infrared absorption arises primarily from oxy and deoxy hemoglobin so:

$$\Delta\mu_a(\lambda) = \epsilon_{HbO}(\lambda)\Delta[HbO] + \epsilon_{HbR}(\lambda)\Delta[HbR]$$

Where

ϵ_{HbO} is the extinction factor of HbO

ϵ_{HbR} is the extinction factor of HbR

$\Delta[HbO]$is the change in concentration of HbO

$\Delta[HbR]$ is the change in concentration of HbR

Thus the MBLL for a single wavelength is

$$\Delta A(\lambda, L) = d \cdot DPF(\lambda) \cdot (\epsilon_{HbO}(\lambda)\Delta[HbO] + \epsilon_{HbR}(\lambda)\Delta[HbR])$$

Because we have two unknowns Δ[HbO] and Δ[HbR], at least two independent measurements at different wavelengths are required. Stacking equations for chosen wavelengths λ_1 and λ_2 gives:

$$\begin{bmatrix} \Delta A(\lambda_1, L) \\ \Delta A(\lambda_{,2} L) \end{bmatrix} = d \cdot \begin{bmatrix} DPF(\lambda_1)\cdot(\epsilon_{HbO}(\lambda_1)] + \epsilon_{HbR}(\lambda_1)) \\ DPF(\lambda_2)\cdot(\epsilon_{HbO}(\lambda_2) + \epsilon_{HbR}(\lambda_2)) \end{bmatrix} \cdot \begin{bmatrix} \Delta[HbO] \\ \Delta[HbR] \end{bmatrix}$$

Inverting this system recovers relative changes in HbO and HbR concentrations. The condition number of the extinction coefficient matrix dictates stability; this explains why wavelength choice is critical. In summary, MBLL provides a tractable bridge between the rigorous but complex diffusion model of light transport and the practical need for fast, non-invasive monitoring of brain hemodynamics. It leverages the exponential foundation of Beer Lambert's law, corrects it with DPF to handle scattering, and expresses results in a form compatible with real-time signal processing.

Therefore this MBLL matrix equation is the analytical endpoint. In practice as mentioned above the extinction coefficient matrix is inverted to estimate the underlying hemoglobin concentration changes from measured optical density changes but this conclusion is sufficient to support the upcoming section.

4.0 Linking MBLL to EEG:

The EEG and fNIRS modalities capture different but causally related aspects of neural activity. EEG records electrophysiological signals synchronous postsynaptic potentials in pyramidal neurons with millisecond precision. fNIRS, on the other hand, detects slower hemodynamic changes: increased [HbO] and decreased [HbR] that follow neural activation through neurovascular coupling.

(MBLL) we derived is the mathematical backbone that transforms raw optical density changes into hemoglobin concentration dynamics and we can directly map fNIRS signals to physiological states

$$\Delta A(\lambda, t) = d \cdot DPF(\lambda) \cdot (\epsilon_{HbO}(\lambda)\Delta[HbO](t) + \epsilon_{HbR}(\lambda)\Delta[HbR](t))$$

This mapping thus allows us to link EEG to fNIRS through two major frameworks: General linear model integration that we partially explored and state space models that introduces a latent neural drive variable that could potentially produce hybrid interpretable signals.

5.0 Critical evaluation:

Despite its high usability, MBLL carries limitations that are important to take into consideration:

1. **Linearity assumption**
   MBLL relies on the log linearization of BeerLambert and assumes small absorption changes. However large hemodynamic shifts may violate this linearity, leading to inaccuracies.
2. **Pathlength uncertainty**
   The DPF depends on tissue composition, head size and age. Approximating it as constant introduces systematic errors and would complicate the implementation of this in a physics device.
3. **Superficial contamination**
   fNIRS signals include contributions from scalp and skull vasculature. MBLL does not intrinsically separate cortical from extracerebral absorption, potentially confounding results.
4. **Scattering simplification**
   Standard MBLL assumes scattering is static, though as argued above, scattering may contain physiologically relevant dynamics.
5. **Fusion complexity with EEG**
   While MBLL bridges the modalities EEG and fNIRS differ dramatically in temporal and spatial resolution. Alignment requires careful modeling of hemodynamic delays and source localization challenges.

6.0 New theoretical extension

The MBLL assumes scattering remains constant during measurements, treating only absorption changes as dynamic. However, emerging evidence suggests this assumption is incomplete:

- Neuronal swelling during depolarization alters local refractive indices.
- Vascular dilation during hemodynamic responses modifies scattering structures.

This implies that scattering may vary dynamically on neural and vascular timescales which is why an extension to the above research can be the inclusion of an additional time varying scattering term:

$$\Delta A(\lambda, t) = d \cdot DPF(\lambda) \cdot (\epsilon_{HbO}(\lambda)\Delta[HbO](t) + \epsilon_{HbR}(\lambda)\Delta[HbR](t)) + \Delta S(\lambda, t)$$

where $\Delta S(\lambda, t)$ captures scattering related changes that are not due to hemoglobin absorption. However the source of this scattering is yet to be deciphered mathematically.

Potential avenues for this extension include - **(a) multi-wavelength CW-fNIRS**: Deciphering absorption vs. scattering using spectral signatures. **(b) frequency-domain NIRS**: utilizing modulation depth and phase shift that provide independent and detectable sensitivity to scattering. **(c) EEG-informed modeling**: train a model to align transient scattering fluctuations with electrophysiological events through extensive real data simulation and collection.

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